

REPORT ON WEATHER DATA ANALYSIS

1. Introduction

Weather data analysis plays a crucial role in understanding climate patterns, seasonal variations, and environmental changes. By analyzing historical weather data, analysts can identify trends in temperature, precipitation, humidity, and other meteorological factors. These insights can support better decision-making in sectors such as agriculture, disaster management, transportation, and urban planning.

In this project, historical weather data is analyzed using Python libraries such as NumPy, Pandas, and Matplotlib. The analysis focuses on cleaning and preparing the dataset, performing statistical analysis, and visualizing weather patterns over time. The goal is to explore patterns such as temperature changes across months, precipitation trends, seasonal variations, and extreme weather conditions.

The project demonstrates how data analysis techniques can transform raw weather data into meaningful insights through statistical analysis and visualizations.

2. Objectives

The main objectives of this project are:

- To load and inspect historical weather data using Pandas.
- To clean the dataset by handling missing values and converting date columns into appropriate formats.
- To perform descriptive statistical analysis using NumPy, including calculating mean, median, and standard deviation for temperature, humidity, and precipitation.
- To analyze monthly and seasonal weather trends using Pandas group-by operations.
- To visualize weather patterns using Matplotlib, including temperature trends, precipitation comparisons, and temperature distributions.
- To identify extreme weather events by detecting unusually high or low temperature and precipitation values using NumPy filtering techniques.

3. Dataset Description

- **Date:** The recorded date of the weather observation
- **Temperature:** The measured air temperature for the given date
- **Humidity:** The percentage of moisture present in the air on a given day
- **Precipitation:** The amount of rainfall or other forms of water (such as snow) recorded on that date.

4. Data Cleaning & Preprocessing

The dataset was first loaded using the **Pandas read_csv()** function and inspected using **head()**, **info()**, and **describe()** to understand its structure and summary statistics. The dataset contains **1,096 records** with four columns: Date, Temperature, Humidity, and Precipitation.

Missing values were checked using **isnull().sum()**, and no missing values were found. The **Date column**, initially stored as an object (string), was converted into **datetime format** using **pd.to_datetime()** to enable time-based analysis.

After preprocessing, the dataset contained clean and properly formatted data, making it ready for further analysis and visualization.

5. Exploratory Data Analysis & Insights

5.1 Descriptive Statistics of Weather Variables

Findings:

- **Average Temperature:** 9.91 °C
- **Median Temperature:** 9.06 °C
- **Temperature Standard Deviation:** 11.74
- **Average Humidity:** 60.85%
- **Median Humidity:** 60.48%
- **Humidity Standard Deviation:** 17.37
- **Average Precipitation:** 3.47 mm
- **Median Precipitation:** 1.52 mm
- **Precipitation Standard Deviation:** 7.40

Insight:

- Temperature shows high variability, indicating significant seasonal changes.
- Humidity remains relatively stable around the 60% range.
- Precipitation varies widely, suggesting occasional heavy rainfall events.

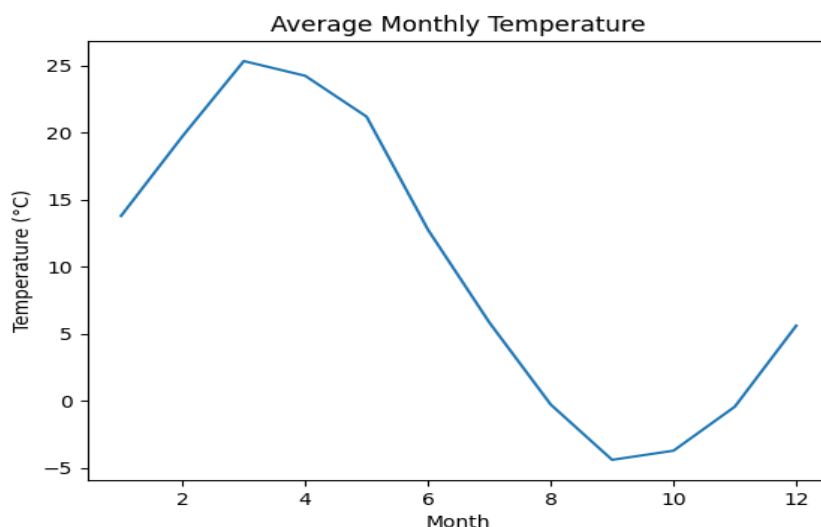
5.2 Monthly Temperature Trend

Findings:

- Temperatures gradually increase from January to March.
- Peak temperatures occur during March and April.
- A decline in temperature begins around June.
- The lowest temperatures are observed between August and October.

Insight:

The dataset shows clear seasonal temperature variation, with warmer conditions in early months and cooler temperatures later in the year.



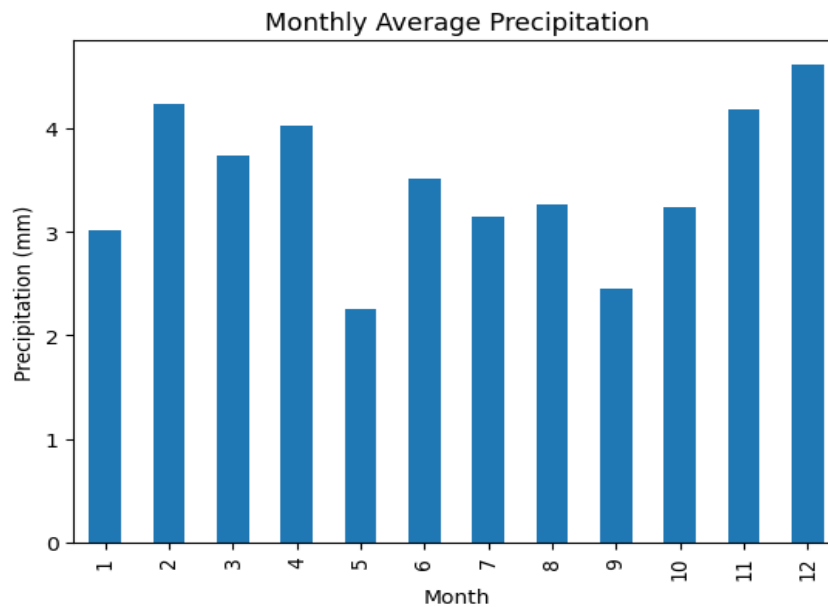
5.3 Monthly Precipitation Pattern

Findings:

- Higher precipitation levels are observed in February, April, November, and December.
- Lower precipitation levels occur in May and September.

Insight:

Rainfall distribution varies across months, indicating periods of heavier rainfall and relatively dry months.



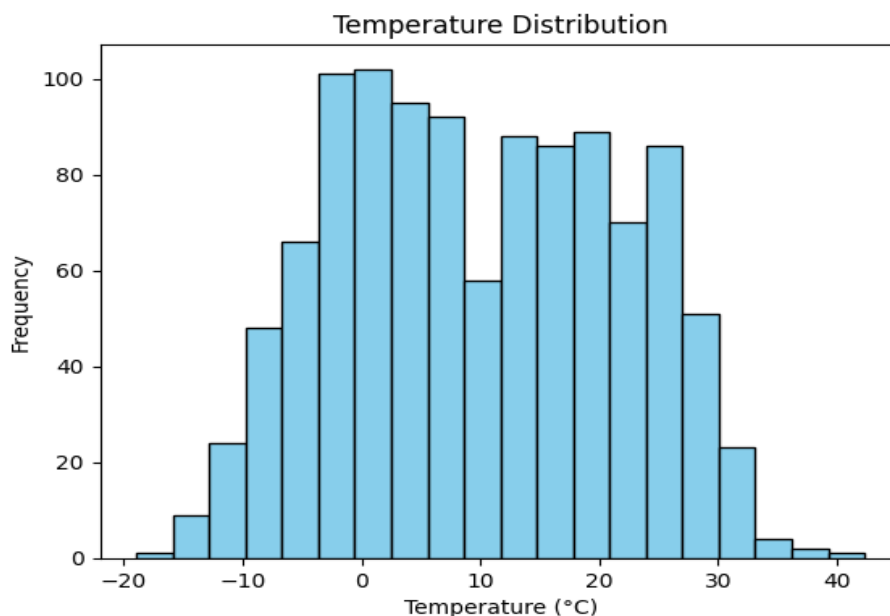
5.4 Temperature Distribution

Findings:

- Temperature values are spread across a wide range from -18.9 °C to 42.3 °C.
- Most temperature observations fall within the moderate temperature range.

Insight:

The histogram shows a broad distribution of temperatures, reflecting both cold and warm periods within the dataset.



5.5 Extreme Temperature Events

Findings:

- Several days recorded extremely high temperatures above 34 °C.
- Some days recorded extremely low temperatures below -14 °C.

Insight:

Extreme temperature events occur occasionally and may represent unusual weather conditions or seasonal extremes.

5.6 Extreme Precipitation Events

Findings:

- A small number of days experienced very high precipitation levels, exceeding the 95th percentile.
- Some rainfall values exceeded 50 mm, indicating heavy rainfall events.

Insight:

Although most days have low to moderate precipitation, the dataset includes occasional heavy rainfall events, which could indicate storms or extreme weather conditions.

6. Key Insights

- The dataset shows clear seasonal temperature variation, with higher temperatures during the early months of the year and lower temperatures between August and October.
- The average temperature is around 9.9 °C, but the large standard deviation indicates significant fluctuations across different seasons.
- Humidity levels remain relatively stable, averaging around 60%, suggesting moderate atmospheric moisture throughout the observed period.
- Precipitation values vary widely, with most days experiencing low rainfall but a few days recording extremely high precipitation.
- The monthly precipitation analysis indicates that some months experience higher rainfall, highlighting uneven rainfall distribution across the year.
- Extreme weather analysis identified several unusually hot and extremely cold days, as well as heavy rainfall events exceeding the 95th percentile.

7. Recommendations

- **Monitor seasonal temperature patterns** to better prepare for extreme weather conditions during peak summer and winter months.
- **Implement early warning systems** for extreme temperature and heavy rainfall events to reduce potential environmental and societal impacts.
- **Improve rainfall monitoring and forecasting**, especially during months with higher precipitation, to support better water resource management.
- **Use historical weather trends for predictive analysis**, which can help forecast future weather patterns and assist in planning for agriculture, transportation, and disaster management.
- **Collect and analyze additional weather variables** such as wind speed, air pressure, and cloud cover to gain deeper insights into climate behavior.
- **Develop automated weather dashboards** using tools like Power BI or Python visualization libraries to monitor weather trends in real time.

8. Conclusion

This project analyzed historical weather data using Python, NumPy, Pandas, and Matplotlib to identify patterns and trends in temperature, humidity, and precipitation. The dataset was cleaned and prepared to ensure accurate analysis, followed by statistical calculations and visualizations to better understand weather behavior over time.

The analysis revealed clear seasonal temperature variations, relatively stable humidity levels, and uneven precipitation distribution throughout the year. Additionally, several extreme temperature and heavy rainfall events were identified, highlighting the variability in weather conditions.

Overall, this project demonstrates how data analysis techniques can transform raw weather data into meaningful insights. Such insights can support better planning and decision-making in areas such as climate monitoring, agriculture, and disaster preparedness.